Formalizing Heterogeneous Interaction Modeling for Multi-Agent Debate in Visual Geo-Localization

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***Abstract*—Visual geo localization requires extensive geographic knowledge and complex reasoning to infer image locations without GPS metadata. Traditional retrieval methods rely on matching to geo tagged databases and performance remains constrained by database coverage and quality. Recent Large Vision Language Models LVLMs enable direct location reasoning from image content but individual models still struggle with diverse geographic regions and complex scenes. Existing multi agent systems improve performance through model collaboration yet they treat all agent interactions in a uniform way and lack mechanisms for effective resolution of conflicting predictions. We propose GraphGeo, a multi agent debate framework based on heterogeneous graph neural networks for visual geo localization. The framework models diverse debate relationships through typed edges and separates supportive collaboration, competitive argumentation and knowledge transfer. We introduce a dual level debate mechanism that combines node level refinement with edge level modeling of argumentation states. A cross level topology refinement strategy supports joint evolution of graph structure and agent representations. Experiments on multiple benchmarks show that GraphGeo outperforms strong single model and multi agent baselines. The framework converts cognitive conflicts between agents into higher geo localization accuracy through structured debate.**

***Index Terms*—Large Visual Geo localization, Multi agent, Debate, Heterogeneous Graph Neural Networks, Large Vision Language Models**

I. Introduction

Visual geo localization aims to identify the geographic position of an image without GPS metadata. This task plays an important role in applications that include analysis of historical human mobility, support for autonomous navigation and location aware recommendation services [1], [2]. Traditional methods address this problem by retrieving similar images from large geo tagged databases [3], [4]. Performance remains limited by database scope, update frequency and record accuracy. Recent progress in Large Vision Language Models LVLMs opens new opportunities and enables direct location reasoning from image content [5], [6]. These models excel in interpreting complex visual information and absorb rich geospatial knowledge during pre training on diverse multimodal data.

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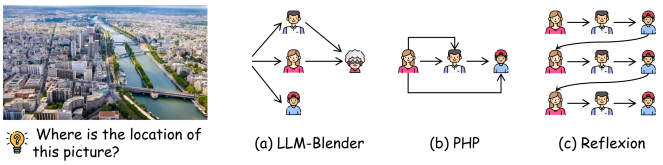


Fig. 1: Comparison of multi-agent collaboration frameworks for visual geo-localization.

Despite these advances, individual LVLMs still face challenges in complex geo localization scenarios [6]. A single model can struggle with fine grained features across culturally diverse regions and can favor training locations that appear more often in pre training corpora. The same model often produces inconsistent predictions in the presence of ambiguous visual cues or highly specialized geographic scenes. To mitigate these issues, researchers have proposed multi agent collaboration frameworks that combine several LVLMs with complementary strengths [7]–[9]. Current systems usually adopt independent exploration with simple result aggregation or sequential collaboration with a fixed interaction order [6]. These strategies leverage model diversity for improved performance but they still lack robust mechanisms for resolution of conflicting predictions. Simple aggregation cannot assess prediction reliability or merge competing interpretations in a principled way. As a result, existing approaches miss the chance to use structured argumentation to expose reasoning flaws and refine predictions through iterative feedback [10].

A key technical challenge is to model the heterogeneous nature of agent interactions and to exploit this structure for conflict resolution. Different agents form distinct relationships based on prediction consistency and confidence. Agents with similar predictions benefit from supportive collaboration that consolidates shared evidence. Agents with conflicting predictions need competitive argumentation that challenges assumptions and reveals errors. Real world geo localization often contains such mixed signals since architectural elements can suggest one region and natural landscapes can indicate another location. Simple aggregation methods such as majority voting cannot measure prediction reliability or detect conflicts that reveal complementary evidence. Knowledge needs directional flow from high confidence agents toward uncertain agents.

Human experts handle disagreements through explicit reasoning, presentation of evidence and careful examination of alternative hypotheses. They build consensus through structured debate that exposes flaws and synthesizes diverse perspectives. Many current graph neural networks assume homogeneous connectivity and apply the same message passing rule on all edges so they cannot express distinct modes of cooperation and competition among agents.

Another challenge arises from dynamic evolution of debate relationships during collaboration. Interaction roles such as agreement, conflict and knowledge transfer can change as discussion progresses. Two agents can move from disagreement to consensus when they gather additional evidence. Agents that appear aligned at first can later expose subtle differences that require deeper analysis. Standard graph neural networks usually update node representations but keep edge relationships fixed. As a result they cannot track how argument strength and reliability evolve across debate rounds. Interaction history carries useful signals for design of collaboration patterns. An effective framework needs explicit pairwise modeling of interactions and adaptive graph topology so both node states and edge relations benefit from ongoing debate.

To address these challenges, we propose GraphGeo, a multi agent debate framework for visual geo localization based on a heterogeneous graph neural network. The framework turns cognitive conflicts into useful signals through structured debate among LVLM agents. GraphGeo represents each agent as a node and encodes debate relations as typed edges in a heterogeneous graph. Agreement edges support consolidation of knowledge through mutual reinforcement. Conflict edges drive competitive argument through critical feedback. Transfer edges guide knowledge sharing from confident agents toward uncertain agents. We design a dual level debate mechanism. Node level refinement aggregates relation specific messages through tailored message passing and edge level modeling maintains learnable representations of pairwise argument states. These edge states evolve through recurrent updates and shape later node reasoning. We also introduce cross level topology refinement so graph structure and agent representations co evolve during debate. In summary our work makes three main contributions:

- We formulate multi agent visual geo localization as structured debate on a heterogeneous graph with typed relations for agreement, conflict and knowledge transfer. This formulation provides a unified view of cooperation and competition in LVLM based multi agent systems.
- We develop GraphGeo, an architecture with dual level debate and cross level topology refinement. Node representations and collaboration patterns evolve jointly and convert cognitive conflicts into accurate geo localization predictions.
- Extensive experiments show that heterogeneous debate improves both accuracy and robustness over strong single model and multi agent baselines. This study highlights the importance of structured interaction design for future LVLM collaboration frameworks.

II. Preliminary

A. Problem Formulation

Visual geo-localization aims to predict the geographic location $\mathcal{Y}$ of a given image $\mathcal{X}$ without relying on GPS metadata. We formulate this task within a multi-agent framework where N large vision-language models serve as collaborative agents $\mathcal{V} = \{v_1, v_2, \ldots, v_N\}$. Each agent v_i generates location predictions $\mathcal{Y}_{v_i}$ with confidence scores $c_{v_i} \in [0, 1]$. The collaborative prediction process can be modeled as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where nodes represent agents and edges $\mathcal{E}$ encode their interaction patterns. The optimal prediction $\mathcal{Y}^*$ emerges from aggregating diverse agent perspectives weighted by their expertise and mutual influence within the graph structure.

B. Graph Neural Networks for Agent Collaboration

Graph neural networks provide a principled framework for learning representations through iterative message passing. Given the agent collaboration graph $\mathcal{G}$, each agent v_i maintains a hidden state $\mathbf{h}_i^{(l)} \in \mathbb{R}^d$ at layer l. The standard message passing mechanism updates node representations by aggregating neighborhood information through

$$\mathbf{h}_i^{(l+1)} = \text{UPDATE}^{(l)}\left(\mathbf{h}_i^{(l)}, \mathbf{a}_i^{(l)}\right), \tag{1}$$

$$\mathbf{a}_i^{(l)} = \text{AGGREGATE}^{(l)}\left(\{\mathbf{m}_{ji}^{(l)} \mid v_j \in \mathcal{N}(v_i)\}\right). \tag{2}$$

where $\mathcal{N}(v_i)$ denotes the neighbors of agent v_i and $\mathbf{m}_{ji}^{(l)}$ represents the message from agent v_j to v_i at layer l. However, this homogeneous treatment of agent interactions fails to capture the competitive and collaborative dynamics essential for effective geo-localization. Agents may produce conflicting predictions that require resolution through argumentation rather than simple averaging, and different images demand different collaboration patterns among agents.

III. Methodology

A. Heterogeneous Graph Construction with Debate Edges

We construct a heterogeneous agent collaboration graph that explicitly models diverse interaction types through typed edges. The graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R})$ incorporates a set of relation types $\mathcal{R} = \{r_{\text{agree}}, r_{\text{conflict}}, r_{\text{transfer}}\}$ to distinguish between different debate mechanisms. For each image $\mathcal{X}$, agent nodes $v_i \in \mathcal{V}$ have features $\mathbf{x}_i^{(0)}$ that concatenate the VAE-encoded image representation with learnable agent embeddings $\mathbf{e}_i \in \mathbb{R}^{d_e}$. The edge set $\mathcal{E}$ is dynamically constructed based on initial agent predictions. We compute the geodesic distance $d_{\text{geo}}(\mathcal{Y}_{v_i}, \mathcal{Y}_{v_j})$ between predicted locations and assign relation types through

$$r_{ij} = \begin{cases} r_{\text{agree}} & \text{if } d_{\text{geo}}(\mathcal{Y}_{v_i}, \mathcal{Y}_{v_j}) < \tau_{\text{agree}} \\ r_{\text{conflict}} & \text{if } d_{\text{geo}}(\mathcal{Y}_{v_i}, \mathcal{Y}_{v_j}) > \tau_{\text{conflict}} \\ r_{\text{transfer}} & \text{if } c_{v_i} - c_{v_j} > \tau_{\text{conf}} \end{cases}, \tag{3}$$

where τ_{agree}, τ_{conflict}, and τ_{conf} are threshold hyperparameters. Agreement edges connect agents with consistent predictions to facilitate knowledge consolidation, conflict edges link agents

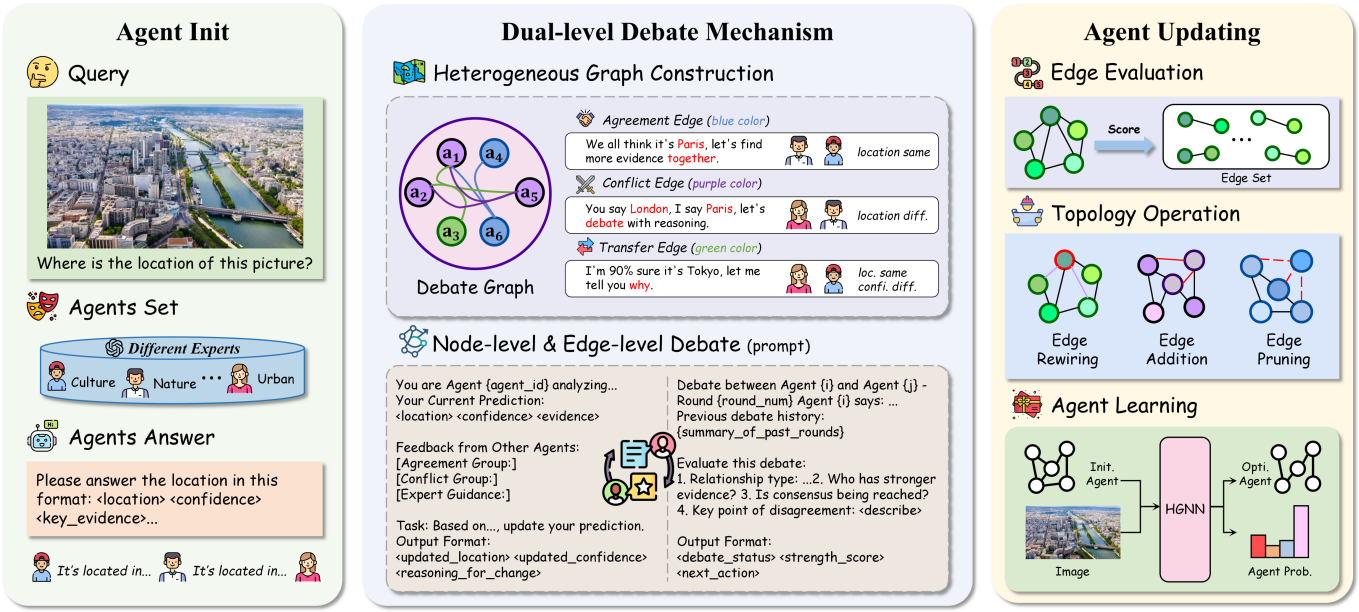


Fig. 2: The framework overview of our proposed method. It contains a dual-level debate mechanism and cross-level topology refinement for visual geo-localization. The first part constructs a heterogeneous graph with three edge types to model diverse agent interactions. The second part employs node-level and edge-level debate to refine predictions through structured argumentation. The third part optimizes graph topology by evaluating edge utility and adjusting connections during multi-round debates.

with divergent views to enable competitive debate, and transfer edges direct information flow from high-confidence to low-confidence agents.

B. Competitive Message Passing with Node-level Debate

At each layer l, we replace standard message aggregation with a competitive debate mechanism. For each agent v_i, we partition its neighbors $\mathcal{N}(v_i)$ according to edge types as $\mathcal{N}_r(v_i) = \{v_j \mid (v_j, v_i, r) \in \mathcal{E}\}$ for $r \in \mathcal{R}$. The node-level debate generates relation-specific messages tailored to the interaction type. For agreement edges, agent v_j generates supportive messages that reinforce shared understanding through

$$\mathbf{m}_{ji}^{\text{agree}} = \mathbf{W}_{\text{agree}} \cdot \sigma(\mathbf{h}_j^{(l)} \odot \mathbf{h}_i^{(l)}), \tag{4}$$

where $\mathbf{W}_{\text{agree}} \in \mathbb{R}^{d \times d}$ is a learnable weight matrix and $\odot$ represents element-wise multiplication. For conflict edges, agent v_j generates critical messages that challenge the current hypothesis through

$$\mathbf{m}_{ji}^{\text{conflict}} = \mathbf{W}_{\text{conflict}} \cdot \text{ReLU}(\mathbf{h}_j^{(l)} - \mathbf{h}_i^{(l)}) \tag{5}$$

Transfer edges employ an attention-based mechanism where high-confidence agent v_j selectively transfers relevant knowledge through

$$\mathbf{m}_{ji}^{\text{transfer}} = \alpha_{ji} \mathbf{W}_{\text{transfer}} \mathbf{h}_j^{(l)}, \tag{6}$$

$$\alpha_{ji} = \frac{\exp(c_{v_j} \cdot \mathbf{h}_j^{(l)\top} \mathbf{W}_{\text{attn}} \mathbf{h}_i^{(l)})}{\sum_{v_k \in \mathcal{N}_{\text{transfer}}(v_i)} \exp(c_{v_k} \cdot \mathbf{h}_k^{(l)\top} \mathbf{W}_{\text{attn}} \mathbf{h}_i^{(l)})} \tag{7}$$

where α_{ji} weights the transfer by both confidence c_{v_j} and semantic relevance. The agent v_i then aggregates these heterogeneous messages through relation-aware pooling

$$\mathbf{m}_i^{(l)} = \bigoplus_{r \in \mathcal{R}} \text{POOL}\left(\{\mathbf{m}_{ji}^r \mid v_j \in \mathcal{N}_r(v_i)\}\right), \tag{8}$$

where $\bigoplus$ denotes a learnable combination operator that balances different debate mechanisms.

C. Edge-level Debate through Cross-agent Interaction

Beyond node-level refinement, we introduce edge-level debate that explicitly models pairwise interactions to capture fine-grained argumentation dynamics. For each edge $(v_i, v_j, r) \in \mathcal{E}$, we construct a debate representation $\mathbf{d}_{ij}^{(l)}$ that encodes the interaction state between agents. This edge representation evolves through

$$\mathbf{d}_{ij}^{(l)} = \mathbf{W}_{\text{edge}}^r \cdot [\mathbf{h}_i^{(l)} \oplus \mathbf{h}_j^{(l)} \oplus \Delta\mathcal{Y}_{ij}], \tag{9}$$

$$\mathbf{d}_{ij}^{(l+1)} = \text{GRU}\left(\mathbf{d}_{ij}^{(l)}, \mathbf{h}_i^{(l+1)} + \mathbf{h}_j^{(l+1)}\right), \tag{10}$$

where $\mathbf{W}_{\text{edge}}^r \in \mathbb{R}^{d_e \times 3d}$ is a relation-specific projection matrix and GRU maintains debate history while incorporating updated node states. The edge representations subsequently influence node updates through edge-to-node message passing

$$\mathbf{m}_i^{\text{edge},(l)} = \sum_{v_j \in \mathcal{N}(v_i)} \beta_{ij} \mathbf{d}_{ij}^{(l)}, \tag{11}$$

$$\beta_{ij} = \text{softmax}_j(\mathbf{d}_{ij}^{(l)\top} \mathbf{W}_{\text{attend}} \mathbf{h}_i^{(l)}) \tag{12}$$

The final node update integrates both node-level and edge-level messages through a gated fusion mechanism

$$\mathbf{h}_i^{(l+1)} = \text{GRU}\left(\mathbf{h}_i^{(l)}, \gamma_i \mathbf{m}_i^{(l)} + (1-\gamma_i)\mathbf{m}_i^{\text{edge},(l)}\right), \quad (13)$$

where the gate $\gamma_i = \sigma(\mathbf{w}_\gamma^\top [\mathbf{h}_i^{(l)} \oplus \mathbf{m}_i^{(l)} \oplus \mathbf{m}_i^{\text{edge},(l)}])$ adaptively balances the two information sources.

D. Cross-level Topology Refinement

To enable dynamic adaptation of the debate structure, we introduce a cross-level topology refinement mechanism. After L layers of message passing, we obtain final agent representations $\mathbf{H} = \{\mathbf{h}_i^{(L)} \mid v_i \in \mathcal{V}\}$ that encode refined predictions. These representations feedback to the graph topology through a learned edge scoring function

$$s_{ij} = \text{MLP}_{\text{score}}\left([\mathbf{h}_i^{(L)} \oplus \mathbf{h}_j^{(L)} \oplus \mathbf{d}_{ij}^{(L)} \oplus \Delta\mathbf{h}_{ij}]\right), \quad (14)$$

where $\Delta\mathbf{h}_{ij} = \mathbf{h}_i^{(L)} - \mathbf{h}_i^{(0)}$ captures the representational change induced by debate. Edges with low utility scores are pruned while new edges are added between agents whose representations become similar after debate. The topology learning objective combines task-specific loss with graph regularization

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{geo}} + \lambda_{\text{sparse}} \sum_{(i,j)\in\mathcal{E}} s_{ij} + \lambda_{\text{diverse}} \mathcal{H}(\mathcal{R}), \quad (15)$$

where $\mathcal{L}_{\text{geo}}$ measures the geodesic distance between predicted and ground-truth locations, the sparsity term penalizes excessive edge connectivity, and $\mathcal{H}(\mathcal{R})$ represents the entropy over relation type distribution. This cross-level interaction ensures that the topology evolves to support effective collaboration patterns while node representations adapt to leverage the refined debate structure.

IV. Experiments

A. Datasets

We evaluate our approach on four geo-localization benchmarks. The **IM2GPS3K** dataset [11] contains 3,000 images sampled from the original IM2GPS collection, covering multiple continents with urban, natural, and landmark scenes. The **YFCC4K** dataset [11] consists of 4,536 geo-tagged images from Yahoo Flickr Creative Commons 100 Million collection with broad visual and geographic diversity. We also utilize the **GeoGlobe** dataset [6], divided into **GeoGlobe (Natural)** for natural attractions and **GeoGlobe (ManMade)** for man-made landmarks. This dataset filters out ambiguous images and provides clearer localization targets across different regions.

B. Baselines

We compare our proposed method with three categories of baselines. First, we evaluate **single LVLM agents** including both open-source models (Qwen-VL [12], Qwen2-VL [13], InternVL2 [14], LLaVA series [15], CogVLM [16]) and proprietary models (Claude-3.5-Sonnet [17], GPT-4 series [18], Gemini series [19]) to establish individual model performance. Second, we include **traditional visual geo-localization methods** such as NetVLAD [3], which pioneered trainable VLAD pooling, GeM [4] using generalized mean pooling, CosPlace [20] reformulating the task as classification for scalability, PIGEON [21], StreetCLIP [22], and GeoCLIP [5] integrating semantic reasoning into features. Third, we compare with **smileGeo** [6], a swarm intelligence framework aggregating multiple LVLM agents through collaborative reasoning, representing the current state-of-the-art in multi-agent geo-localization.

TABLE I: Results of different single LVLM baselines and comparison with baselines. Bold indicates the statistically significant improvements over the best baseline.

Model	IM2GPS3K	YFCC4K	GeoGlobe (Natural)	GeoGlobe (ManMade)
Open-source LVLMs				
Infi-MM	14.7000	6.5256	19.2020	21.4145
Qwen-VL	32.4667	14.3959	42.3940	37.4556
Qwen2-VL-7B	36.2333	15.8724	45.6783	52.3142
InternVL2-8B	33.5667	14.9213	40.1256	48.7623
vip-llava-13b	15.7667	6.9885	20.6983	15.4089
vip-llava-7b	38.7000	17.8792	31.9202	56.4994
llava-1.5-7b	30.2000	13.8889	27.1820	47.2145
llava-1.6-7b-mistral	3.1000	1.6314	0.7481	2.1731
llava-1.6-7b-vicuna	9.4667	4.4092	6.9825	15.8831
llava-1.6-13b	17.8333	8.3333	12.2195	28.2497
llava-1.6-34b	44.2667	20.8113	52.8678	77.2027
CogVLM	6.7333	3.0644	7.7307	10.3516
CogVLM2	28.4333	12.5678	35.4521	42.3187
Proprietary LVLMs				
Claude-3-Opus	23.8333	12.3457	33.1671	40.6954
Claude-3.5-Sonnet	41.2667	18.6735	55.3214	76.8456
GPT-4V	43.5333	19.2147	58.4672	80.1234
GPT-4o	46.8333	20.4569	64.7823	85.2617
GPT-4o-mini	45.0000	18.3422	62.0948	84.5911
Gemini-1.5-Flash	39.4667	17.2341	54.8912	74.5623
Gemini-1.5-Pro	47.3667	19.9956	62.3441	82.8131
Visual Geo-localization Methods				
NetVLAD	16.6303	7.4876	26.5134	28.9955
GeM	14.4907	6.5243	23.1022	25.4175
CosPlace	17.6686	7.9551	28.1688	30.2782
PIGEON	29.5632	12.8914	35.7891	41.2365
StreetCLIP	33.2145	14.6782	39.5621	44.8912
GeoCLIP	34.4728	15.1719	38.2519	45.9174
smileGeo	47.7667	21.5168	76.0599	85.4603
Ours	**49.1456**	**22.1623**	**77.8217**	**87.9341**

C. Metrics

Following common practice in visual geo localization, we adopt street level accuracy as the main evaluation metric. It measures the percentage of images localized within a distance threshold from the ground truth. We report accuracy at several scales from fine street level at 1 km to coarse continent level at 2500 km. For each dataset, we compute the proportion of test images whose predicted locations lie inside the given radius around the true coordinates. This metric is easy to interpret and supports fair comparison across methods and datasets.

D. Implementation Details

We implement all experiments on NVIDIA A100 GPUs. Our multi-agent system incorporates a diverse set of LVLM agents selected from both open-source and proprietary models

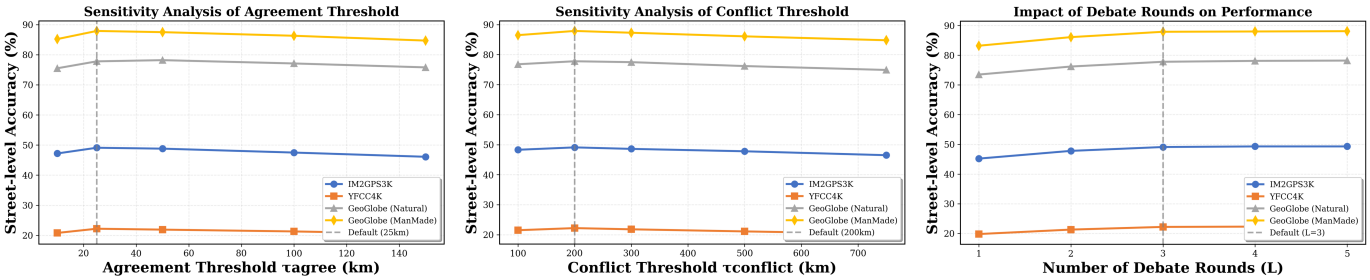


Fig. 3: Hyperparameter sensitivity analysis. (a) Agreement threshold τ_{agree} shows optimal values vary across datasets due to different geographic distributions, with default at 25 km. (b) Conflict threshold $\tau_{conflict}$ balances critical debate and noise, where lower values create more conflict edges. (c) Debate rounds show performance gains diminish after 3 rounds, making $L = 3$ optimal for accuracy-cost trade-off.

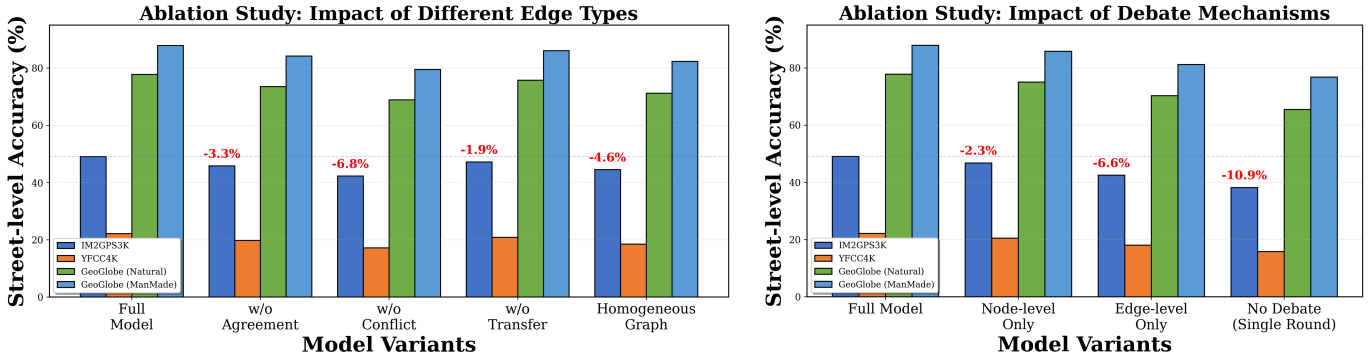


Fig. 4: Ablation Study on Edge Types and Debate Mechanism Caption: Ablation studies on heterogeneous edge types and dual-level debate mechanism. Conflict edges are most critical, with 6.8% performance drop when removed, highlighting the essential role of competitive debate in error correction. Node-level aggregation proves more critical than edge-level modeling, while removing the entire debate mechanism causes 10.9% performance drop on IM2GPS3K, validating the core value of iterative multi-agent argumentation.

to ensure complementary reasoning capabilities. For graph construction, we initialize the heterogeneous graph with typed edges based on prediction similarity, setting distance thresholds $\tau_{\text{agree}} = 25\text{km}$, $\tau_{\text{conflict}} = 200\text{km}$, and $\tau_{\text{conf}} = 0.3$ for edge type assignment. The dual-level debate mechanism operates through $L = 3$ layers of message passing with hidden dimension $d = 512$. We employ the Adam optimizer with learning rate 1×10^{-4} and train for 50 epochs with batch size 32. The cross-level topology refinement uses hyperparameters $\lambda_{\text{sparse}} = 0.01$ and $\lambda_{\text{diverse}} = 0.05$ to balance prediction accuracy with graph structure quality. During inference, we aggregate final agent predictions through confidence-weighted voting, where each agent's contribution is determined by its refined confidence score after multi-round debate.

E. Main Results and Model Analysis

Table I presents the results of our method and all baselines including open source LVLMs, proprietary models, traditional geo localization methods, and the multi agent system smileGeo. Our method attains the best performance on all four benchmarks with street level accuracy of 49.1% on IM2GPS3K and 87.9% on GeoGlobe ManMade and it surpasses the strongest baseline smileGeo on both IM2GPS3K and GeoGlobe by clear margins. Proprietary LVLMs show strong single model ability and open source LVLMs display larger variance in accuracy and traditional geo localization methods remain far behind vision language models, so semantic reasoning becomes a key factor for this task. The multi agent baseline smileGeo already improves over every single model through collaborative reasoning yet Our method brings further gains. The core idea behind Our method is that interactions among agents are heterogeneous and need different handling through typed edges. Agreement edges support collaboration and knowledge consolidation, conflict edges drive competitive argument and error discovery, and transfer edges guide knowledge flow from confident agents to uncertain agents. The dual level debate mechanism combines node level message aggregation with edge level interaction modeling so agents can absorb diverse views and keep explicit debate states that evolve across rounds.

F. Hyper-parameter Analysis

Figure 3 examines the sensitivity of key hyperparameters across our four benchmarks. The agreement threshold shows optimal values vary by dataset, with 25km working best overall as it balances supportive collaboration without over-connecting

dissimilar predictions. Too small values create overly sparse agreement graphs that limit beneficial consolidation. The conflict threshold at 200km effectively separates genuine disagreements from noise. Too large values blur the distinction between conflicting and merely distant predictions. Debate rounds show performance gains through three iterations, after which improvements plateau as agents reach consensus. The dynamic nature of our graph topology allows the framework to adapt edge connections based on debate effectiveness, with cross-level refinement enabling bidirectional influence between structure and representations. This adaptive mechanism ensures that productive debate relationships strengthen over time while unproductive connections are pruned.

G. Ablation Study

Figure 4 systematically evaluates each component's contribution to our framework. Removing conflict edges causes the largest performance drop of 6.8% on IM2GPS3K, highlighting that competitive argumentation is essential for error correction. Agreement edges contribute 3.3% and transfer edges add 1.9%, while the homogeneous graph variant loses 4.6% by treating all interactions uniformly. Eliminating all iterative refinement causes a dramatic 10.9% performance drop, validating that multi-round argumentation forms the core value of our approach. The results reveal that different agent relationships serve fundamentally different purposes in collaborative geo-localization. Agreement requires consolidation of shared understanding, conflict requires structured channels for challenging assumptions, and confidence gaps require directional knowledge flow. Our heterogeneous graph design captures these distinct interaction semantics, enabling agents to leverage both cooperation and competition.

V. Conclusion

This paper introduces GraphGeo, a multi-agent debate framework for visual geo-localization using heterogeneous graph neural networks. By modeling diverse agent interactions as typed edges and implementing dual-level debate with cross-level topology refinement, our framework turns cognitive conflicts into enhanced localization accuracy. Results demonstrate that heterogeneous interaction modeling and structured debate are crucial for multi-agent consensus under uncertainty.

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